

Psychometric properties of a medium version of the Copenhagen Psychosocial Questionnaire (COPSOQ II) for southern Brazil

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Abstract.

BACKGROUND: The emerging frequency of Behavioural Mental Health Disorders among Brazilian workers and the recent legal demand for analysis of psychosocial risks in the workplace highlight the importance of standardizing measures to assess these risks as a way to allow identification and proper comparison among different populations.

OBJECTIVE: To assess the psychometric properties of the COPSOQ II questionnaire medium version for southern Brazil, based on the Spanish medium-length version of COPSOQ-ISTAS21 II.

METHODS: A sample of 426 workers from a university in southern Brazil answered the model under study online. Content validity and internal consistency were analyzed through Confirmatory Factor Analysis (AFC) and Exploratory Factor Analysis (AFE) and Cronbach's α coefficient.

RESULTS: The study model presented a response rate of 48.46%. The analyses indicated the possibility of the instrument to present reliability and validity of content. From the AFE, the final model consisted of 13 dimensions and 70 items, and presented a Cronbach's alpha of 0.82, which is considered a good internal consistency.

CONCLUSIONS: The results showed that the final model of this study presents acceptable levels of reliability and internal validity for the application in Brazil, along with the groups of workers that resemble the participants of the research, to assess psychosocial risks in the workplace.

Keywords: Occupational health, Occupational risks, surveys and questionnaires, Psychometry

1. Introduction

Psychosocial risks (related to work organization and working relationships) and their consequences (sleep / musculoskeletal / cardiovascular disorders,

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occupational accidents, mental health disorders), affect the well-being and the health and safety of people in the context of work, and their causes may concern all organizations, regardless of size and sector of activity [1–3].

In 1984, the International Labor Organization (ILO) and the World Health Organization (WHO) defined psychosocial risks in the workplace as the interaction between work environment, work content, organizational conditions and the worker, which can influence health, performance and job satisfaction [4]. This definition emphasized the dynamic interaction between the work environment and human factors [5].

Government and non-governmental organizations in different countries follow guidelines and legislation that support the need to understand more deeply the factors of psychosocial risks at work, considering them as significant in the origin of different health and quality of life of workers. For example, the Global Plan of Action on Workers' Health, published by the WHO in 2007, guided the elaboration of actions with this focus, such as a creation of national policy frameworks for workers' health and of a national action plans on workers' health, between 2008 and 2017 [6].

However, attention to psychosocial factors in developing countries encounters some barriers, such as the conceptual consensus of the variables involved and existing health and safety legislation [7]. The monthly follow-up reports on the Brazilian social sickness benefits, created by the Ministry of Finance's Secretariat of Social Security (SF-MF) since 2006; and the advent of the System of Digital Bookkeeping of Tax, Social Security and Labor Obligations (eSocial) by the federal government may justify the urgent need for reliable instruments that support the assessment of psychosocial risks in the workplace in Brazil. The SF-MF data allow us to verify that medical leave due to Behavioural Mental Health Disorders has been progressively increasing in Brazil's counties; and has been characterized as the 4th reason for withdrawal of work between January and December 2017 [8]. Thus, since 2018 eSocial requires Brazilian employers to provide data on workers' working conditions, including the identification and measurement of exposure to psychosocial risks [9].

Several theoretical approaches have been developed on psychosocial risks and these models have enabled the development of a series of evaluation instruments (surveys and questionnaires, mainly) [3]. Some of the most popular and most used instruments

are the *Job Content Questionnaire* (JCQ), the *General Nordic Questionnaire for Psychosocial and Social Factors at Work* (QPS), the *Effort-Reward Imbalance Questionnaire* (ERI) and the *Copenhagen Psychosocial Questionnaire* (COPSOQ) [10], few of which have been validated for Brazil (JCQ and ERI).

The first version of COPSOQ, developed by the Danish National Institute of Occupational Health in 2000 and published in 2005 [11] presents characteristics that differentiate it from JCQ, QPS and ERI. The COPSOQ is not only based on a theoretical model that explains the relationship between psychosocial risks, work environment and health, but on a systemic approach [10]. This instrument is based on a set of widely accepted theories, such as: (1) Job characteristics model; (2) Michigan organizational stress model; (3) Demand-control model; (4) Sociotechnical approach; (5) Theoretical approach to action and (6) Effort-reward imbalance model [11]. Because of its practical nature, the COPSOQ can include more relevant dimensions for research that may not have been covered in previously validated models, such as JCQ, QPS and ERI; as well as it can include less studied factors [10]. The COPSOQ can be described as a generic tool applicable to all types of workplaces and occupations [6], like private sector administrative workers [12] or hospital employees [13, 14]. Currently, the COPSOQ is one of the most widely used instruments for psychosocial risk assessment in the workplace [15]. It has gained importance in the scientific community in several countries and has been translated into over 25 languages, allowing cross-country comparison [16, 17]. Validation studies were carried out in Germany [18], Spain [19], China [20], France [3], Sweden [21], Chile [22], Romania [23], Iran [24], Portugal [25], among others, in which the COPSOQ achieved good internal consistency and overall Cronbach's alpha above 0.80.

The COPSOQ International Network Steering Committee, an international organization founded in 2009, promotes scientific research and communication among groups that use COPSOQ (governments, universities, researchers and companies). The organization suggests that the use of the instrument in Latin American countries bears on the Spanish version of the questionnaire, the COPSOQ-ISTAS21 II - medium version [19]. The suggestion is justified because it considers that the Spanish version presents a structured scientific methodological procedure for these countries, taking into account the cultural differences and the idiomatic, semantic, conceptual and experimental aspects, and seeking to guarantee

equivalences with the original version, as performed in the validation study for Chile [22].

The emerging frequency of common mental disorders (CMD) cases among Brazilian workers and the recent legal demand for analysis of psychosocial risks in the workplace highlight the importance of standardizing measures to assess these risks as a way to allow identification and proper comparison among different populations. In this context, this study aimed to evaluate the psychometric properties of the COPSOQ II questionnaire medium version for southern Brazil, based on the Spanish version COPSOQ-ISTAS21 II - medium version.

2. Method

This validation study of the medium-length version of COPSOQ-ISTAS21 II for southern Brazil was authorized by the COPSOQ International Network Steering Committee, which guided the use of the version that had already been translated and adapted transculturally to Brazil by the Industry Social Service of Paraná (SESI/PR) [6]. SESI/PR has provided the Sesi Model of Psychosocial Risk Assessment/COPSOQ-ISTAS21 II - medium version (COPSOQ-SesiPR) for the accomplishment of this study.

2.1. Study participants

This cross-sectional study had a target population of 879 workers from a university in the South of Brazil. The inclusion criteria of the study were: to have been an institution worker for at least one year, and to have complete high school.

The sample consisted of 426 individuals (48.46% of the study population), fulfilling the criteria that sample size must be at least five times the number of variables involved in the instrument for an adequate execution of the factorial analysis model [26, 27].

2.2. Instrument

COPSOQ is characterized as an instrument for assessing psychosocial risks in the workplace that (1) defines units of analysis in terciles and allows the existence of a practical assessment and intervention measurement in the workplace; (2) incorporates reference values for the various countries, sectors and occupations in particular; (3) is applicable to all types of organization [10]. It presents three versions: long

(41 dimensions and 128 questions, for research and research purposes), medium (28 dimensions and 87 questions, intended for use by occupational health professionals) and short (23 dimensions and 40 questions for self-evaluation or application in places with less than 30 employees) [6].

All dimensions are considered as differentiated and operationally measured conceptual entities, together they form part of the same psychological construct, and are independent and distinct in their measurement and function in the most diverse realities of the organization and working conditions [19].

All versions of COPSOQ measure the exposure indicators (psychosocial risks) and indicators of their effect (health, satisfaction and stress) and all questions of the three versions are evaluated on a Likert-type scale, with five options of response with a score ranging from 0 to 100, respectively: “Almost Always or Always”, “Often”, “Sometimes”, “Rarely” and “Almost Never or Never”. If the answer to a question is missing, the scale is not calculated [6].

COPSOQ-ISTAS21, Spanish version of the COPSOQ, consists of six domains and 20 dimensions, respectively, as follows: Psychological demands (Quantitative demands, Work pace, Emotional demands, Demands to hide emotions); Work-family conflict (Work-family conflict); Control over work (Influence at work, Possibilities for development, Meaning of work); Social support and quality of leadership (Social support from colleagues, Social support from superiors, Quality of leadership, Social community at work, Predictability, Transparency of labour role, Conflict at work); Compensation at work (Acknowledgment, Job Insecurity, Uncertainty about Working Conditions) and Social Capital (Justice and respect, Vertical trust) [6, 19]. Health, satisfaction and stress dimensions are evaluated by the general health, mental health and vitality scales from the generic SF-36 questionnaire of assessment of quality of life, already validated for Brazil [28].

The dimension “Offensive behaviour” was incorporated into the COPSOQ-SesiPR from the COPSOQ - Portuguese medium version [6, 25], totalling six domains, 21 dimensions and 70 items, as presented in Table 1.

2.3. Data collection procedures

To constitute the sample, the board of directors of the University Campus was contacted by the researchers, who introduced the objectives and methodology of the study. Duly authorized by the

Table 1
COPSOQ-SesiPR domains, dimensions and items – Source: [6]

Domains	Dimensions	Items
Job Demands	Work pace	1; 2
	Emotional demands	3; 4; 5
	Demands to hide emotions	6; 7; 8; 9
	Quantitative demands	10; 11; 12; 13
Work and content organization	Influence at work	14; 15; 16; 17
	The meaning of work	22; 23; 24
	Possibilities for development	25; 26; 27; 28
Social relations and leadership	Predictability	29; 30
	Transparency of labor role	31; 32; 33; 34
	Conflict at work	35; 36; 37; 38
	Social support from colleagues	39; 40; 41
	Social support from superiors	45; 46; 47
	Acknowledgment at work/Reward	53; 54; 55
	Quality of leadership	63; 64; 65; 66
Values in the workplace	Vertical trust	56; 57; 58
	Justice and respect	59; 60; 61; 62
	Social community at work	42; 43; 44
Offensive behaviour	Offensive behaviour	67; 68; 69; 70
Work-individual interface	Work-family conflict	18; 19; 20; 21
	Job insecurity	48; 49; 50
	Job stability	51; 52

board and once the employees' work records had been provided, an email was sent to all those who met the outlined inclusion criteria, presenting the purpose of the study and the electronic link to access the online questionnaire. After reaching the webpage of the questionnaire, the worker read the informed consent form and, upon agreeing, indicated his/her acceptance, which opened the access to the questionnaire. The data was collected between September and December 2017.

The procedures were approved by the local Committee for Ethics in Research with Human Subjects, according to the Declaration of Helsinki.

2.4. Data analysis

The data collected were transcribed in the IBM SPSS Statistics 24 software.

The Kolmogorov-Smirnov test verified the normality of the variables. The result ($p=0.000$ for the 70 items of the questionnaire) found that the variables analysed do not have a normal distribution. To measure the content validity, the Confirmatory Factor Analysis (CFA) and the Exploratory Factor Analysis (EFA) were used, and to analyse the internal consistency of the dimensions of the questionnaire, the Cronbach's α coefficient was used.

Two tests were carried out: the Kaiser-Meyer-Olkin (KMO) test for sampling adequacy measure, for which high values (between 0.5 and 1.0)

indicate that factor analysis is appropriate [29]; and the Bartlett's test of sphericity, which must present significant result ($p \leq 0.05$) to characterize the overall significance of all correlations in a correlation matrix [27]. Then, the slope graphic and Total Variance Explained (TVE) technique of the Principal Component Analysis method was used to extract the factors - the CFA indicated the number of factors suggested by the COPSOQ-ISTAS21 II - medium version [19] and in EFA the model could define the appropriate number of factors. Acceptable commonalities were stipulated in values above 0.60, and the cut-off value for factor extraction was set at 0.50. To achieve a simpler and theoretically more significant factorial pattern, the Varimax-type rotation method was used, with Kaiser normalization.

3. Results

3.1. Sample characterization

The sample consisted of 268 women (62.9%) and 158 men (37.1%), with a median age of 36.73 years (± 11.71) and median working time in the organization of 9.27 years (± 6.82). The most frequent job categories were: administrative auxiliary (107–25.1%), teacher (104–24.4%), administrative assistant (42–9.9%), library auxiliary (12–2.8%) and operational support assistant (10–2.3%). In terms of

Table 2
Number of factors and items resulting from the CFA

Factor	Number of items by dimension of COPSOQ-ISTAS21 II - medium-length version	Items	Total items per factor
1	4 (Quality of leadership) 3 (Acknowledgment at work) 3 (Vertical trust) 4 (Justice and respect) 3 (Social support from superiors) 2 (Predictability)	63; 64; 65; 66 53; 54; 55 56; 57; 58 59; 60; 61; 62 45; 46; 47 29; 30	19
2	2 (Job insecurity) 3 (Uncertainty about working conditions)	51; 52 48; 49; 50	5
3	4 (Possibilities for development)	25; 26; 27; 28	4
4	3 (Offensive behaviour)	68; 69; 70	3
5	4 (Work-family conflict)	18; 19; 20; 21	4
6	3 (Quantitative demands)	11; 12; 13	3
7	2 (Social support from colleagues) 1 (Social community at work)	39; 40 44	3
8	2 (Transparency of labor role)	31; 32	2
9	1 (Conflict at work) 3 (The meaning of work)	38 22; 23; 24	4
10	3 (Conflict at work)	35; 36; 37	3
11	3 (Emotional demands)	3; 4; 5	3
12	2 (Influence at work)	16; 17	2
13	2 (Work pace)	1; 2	2
14	2 (Transparency of labor role)	33; 34	2
15	2 (Social community at work)	42; 43	2
16	2 (Demands to hide the emotions)	8; 9	2
17	1 (Influence at work)	15	1
18	1 (Quantitative demands)	10	1
19	1 (Social support from colleagues)	41	1
20	2 (Demands to hide the emotions)	6; 7	2
21	1 (Influence at work) 1 (Offensive behaviour)	14 67	2
Total Items			70

education, 95 (22.3%) had completed high school, 158 (37.1%) had a complete degree, 39 (9.2%) had specialization, 73 (17.1%) had a master's degree and 61 (14.3%) had a doctoral degree.

3.2. Internal validity

It was found that factor analysis was recommended for the treatment of the data of this study ($KMO = 0.814$) and the Bartlett's test presented a significant result ($p = 0.000$).

3.2.1. Confirmatory factor analysis

To make feasible a comparative analysis, the number of factors considered for the CFA was based on the number of dimensions of the model under study, COPSOQ-ISTAS21 II - medium version [16], consisting of 21 dimensions (Table 1).

In the CFA TVE, 77% of the variance is explained when 21 factors are extracted based on the Kaiser criterion, which predicts to retain factors with an eigenvalue above 1.

The 70 items revealed commonality indexes between 0.635 and 0.916, which is considered acceptable. The analysis of the factorial loads presented in the CFA Rotated Components Matrix (RCM) identified that all had values above 0.40, therefore, no items were found that could be considered without significance and that could be rejected because of that.

Table 2 shows the number of factors and items by dimensions of COPSOQ-ISTAS21 II - medium version [19] resulting from the CFA.

Table 2 presents as outcome of the CFA the extraction of 21 factors. The results allow us to identify that there was a regrouping of items of different dimensions, as seen in factors 1, 2, 7 and 9. Factors 5, 6, 11 and 13 remained constituted by the same items of COPSOQ-ISTAS21 II - medium version [19] factors 4, 10, 12, 15 and 16 had a reduction in the number of items due to the fact that at least 1 item of each of these dimensions constituted another factor.

When compared to COPSOQ-ISTAS21 II - medium version [19], the CFA result with 21 factors presented a difference in the grouping of items, which

Table 3
Number of factors and items resulting from the EFA

Factor	Number of items by dimension of COPSOQ-ISTAS21 II - medium-length version	Items	Total items per factor
1	3 (Quality of leadership) 3 (Acknowledgment at work) 3 (Vertical trust) 4 (Justice and respect) 2 (Social support from superiors) 2 (Predictability) 1 (The meaning of work)	64; 65; 66 53; 54; 55 56; 57; 58 59; 60; 61; 62 45; 46 29; 30 22	18
2	3 (Uncertainty about working conditions) 2 (Job insecurity)	48; 49; 50 51; 52	5
3	1 (Justice and respect) 2 (Quantitative demands) 3 (Emotional demands)	62 11; 13 3; 4; 5	6
4	4 (Possibilities for development)	25; 26; 27; 28	4
5	1 (Predictability) 4 (Transparency of labor role) 1 (The meaning of work)	30 31; 32; 33; 34 24	6
6	4 (Conflict at work) 1 (The meaning of work)	35; 36; 37; 38 23	5
7	4 (Offensive behaviour)	67; 68; 69; 70	4
8	3 (Social support from colleagues) 1 (Social community at work)	39; 40; 41 44	4
9	4 (Work-family conflict)	18; 19; 20; 21	4
10	2 (Influence at work)	16; 17	2
11	4 (Demands to hide the emotions)	6; 7; 8; 9	4
12	2 (Work pace) 1 (Influence at work)	1; 2 14	3
13	2 (Social community at work)	42; 43	2
14	1 (Quantitative demands)	12	1
15	1 (Influence at work)	15	1
16	1 (Quantitative demands)	10	1
Total Items			70

was very heterogeneous. For this reason, we decided to make an EFA to identify how many factors the set of data collected would allow to define.

3.2.2. Exploratory factor analysis

The EFA TVE of the model under study extracted 16 factors which, together, explained 72.02% of the variance, based on the Kaiser criterion. Subsequently, the slope graph analysis was performed, representing that the number of factorial components extracted in relation to the eigenvalues associated to these factors was 16.

The 70 items revealed commonality indexes between 0.608 and 0.896, considered acceptable. The analysis of factorial loads presented in the RCM of the EFA identified that all had values above 0.30, therefore, as in the CFA, no items were found that could be considered without significance and for which they could be rejected.

Table 3 shows the number of factors and items by dimensions of the COPSOQ-ISTAS21 II - medium version resulting from the EFA.

The EFA resulted in the extraction of 16 factors. The results allow us to identify that there was a regrouping of items of different dimensions, as seen in factors 1, 2, 3, 5, 6, 8, e 12. Factors 4, 7, 9 and 11 remained constituted by the same items of COPSOQ-ISTAS21 II – medium version [19], while for factors 10 and 13 the number of items was reduced. Factors 14, 15 and 16 were disregarded because they had only 1 item. The items of the excluded factors were regrouped according to their factorial load in other factors of the study model, since these values do not justify the exclusion of these items. In this way, items 10 (0.688) and 12 (–0.700) were regrouped to factor 3, and item 15 (0.658) was regrouped to factor 10. Thus, EFA resulted in 13 factors, with 70 items.

After the completion of the CFA and the EFA of the model under study, and since the 21 factors of the CFA presented low coincidence with the 21 factors of the COPSOQ-ISTAS21 II - medium version [19], considered more appropriate to work with the factorial grouping resulting from the EFA

Table 4
Internal consistency of Brazilian, Spanish and Danish COPSOQ dimensions

Dimensions	Brazil	Spain	Denmark
	Medium-length Version (n. of items; α)	Medium-length Version (n. of items; α)	Medium-length Version (n. of items; α)
Work pace	2 (0.76)	3 (0.76)	3 (0.84)
Emotional demands	3 (0.77)	4 (0.76)	–
Demands to hide emotions	4 (0.64)	4 (0.62)	–
Quantitative demands	4 (0.92)	4 (0.69)	4 (0.82)
Influence at work	4 (0.57)	4 (0.77)	4 (0.73)
Work-family conflict	4 (0.76)	4 (0.89)	4 (0.80)
The meaning of work	3 (0.72)	3 (0.74)	3 (0.74)
Possibilities for development	4 (0.84)	4 (0.81)	4 (0.77)
Predictability	2 (0.77)	2 (0.59)	2 (0.74)
Transparency of labor role	4 (0.78)	4 (0.78)	–
Conflict at work	4 (0.77)	4 (0.75)	–
Social support from colleagues	3 (0.78)	3 (0.83)	–
Social community at work	3 (0.70)	3 (0.89)	–
Social support from superiors	3 (0.86)	3 (0.87)	4 (0.079)
Uncertainty about working conditions	3 (0.88)	4 (0.78)	–
Uncertainty about the job	2 (0.80)	2 (0.62)	–
Acknowledgement	3 (0.82)	3 (0.79)	3 (0.83)
Vertical trust	3 (0.78)	3 (0.77)	–
Justice and respect	4 (0.86)	4 (0.81)	4 (0.83)
Quality of leadership	4 (0.90)	4 (0.81)	4 (0.89)
Offensive behaviour	4 (0.70)	–	–

(13 factors), to be used for the population of this study, namely: Job management (18 items), Job insecurity (5 items), Demands at work (8 items), Possibility for development (4 items), Transparency of the labour role (6 items) Conflict at work (5 items), Offensive behaviour (4 items), Social relations (4 items), Work-family conflict (4 items), Influence at work (3 items), Demands to hide the emotions (4 items), Work pace (3 items) and Social community at work (2 items).

3.3. Internal consistency of dimensions

Initially, the internal consistency of the 21 dimensions of the questionnaire translated into Brazilian Portuguese (COPSOQ-SesiPR) was analysed by checking the Cronbach's α coefficient, as well as the values of the Spanish [19] and Danish [11] versions, aiming at comparing the three versions (Table 4). Not all dimensions are identical in the three versions, however, the Spanish validation study was conducted based on the medium-length version of the Danish COPSOQ (original version).

The internal consistency of a questionnaire increases the closer to 1 is the value of the statistics analysed. The COPSOQ-SesiPR presented 2 dimensions with excellent internal consistency, 6 with good

Table 5
Internal consistency of the dimensions originated by the EFA of COPSOQ-SesiPR

Dimensions	N° of items	Cronbach's α
Work management	18	0.95
Job insecurity	5	0.89
Demands at work	8	0.81
Possibility for development	4	0.84
Transparency of labor role	6	0.79
Conflict at work	5	0.73
Offensive behaviour	4	0.70
Social relationships	4	0.76
Work-family conflict	4	0.76
Influence at work	3	0.72
Demands to hide the emotions	4	0.64
Work pace	3	0.73
Social community at work	2	0.77

internal consistency, 11 with acceptable internal consistency, 1 with questionable internal consistency and 1 with poor internal consistency, resulting in a general Cronbach alpha of 0.82, classified as good [30].

Considering the EFA results of the COPSOQ-SesiPR, we verified the Cronbach alpha coefficient of the 13 factors extracted renamed (Table 5).

Dimension 1 presented an excellent level of reliability and the other dimensions presented good levels of reliability, resulting in a general Cronbach's alpha of 0.82, classified as a good level of reliability [31].

After the CFA, EFA were carried out and Cronbach's α coefficient of the model under study was verified, the items ($n=70$) and factors ($n=13$) that presented acceptable performances remained, and constituted the final model.

4. Discussion

This study aimed to evaluate the psychometric properties of the Spanish version of the COPSOQ II - medium-length version for the south of Brazil, to identify the items with the best results to compose the version adapted to the Portuguese language spoken in Brazil. For this, the validity and the internal consistency of the instrument were tested, and the results proved that the final model might present acceptable levels of reliability and construct validity. As in the study by Silva, Wendt and Argimon [32], the analyses carried out indicated that it would be convenient to change the study model to reach an acceptable quality, provided they did not violate the COPSOQ theoretical model.

The COPSOQ validation studies generally involve the medium and/or long versions of the questionnaire and are based on different versions of the COPSOQ successively developed by the Danes and other authors [3]. The 70 items that make up the medium-length version of the final model were grouped in 13 dimensions and, with regard to the results of the AFE, are convergent with the Danish and Spanish models, confirming the successive development of the original model of the questionnaire. The dimensions "Work pace", "Influence at work" and "Work-family conflict" are present in all three versions.

It is possible to observe that the alphas derived from the items of the Brazilian version are of the same order of magnitude as the other versions (Tables 4 and 5). The medium and short versions of the Danish questionnaire were not developed focusing only on high alphas but were based primarily on the high content validity of the global instrument [3]. It is known that the value of Cronbach's alpha depends on the number of items involved, tending to be larger with a larger number of items. This feature could explain and justify some weaker alphas (calculated considering only two items) [3, 15]. The difference in the coefficients of internal consistency between dimensions was more evident with respect to the dimension "Job insecurity" of the final model - constituted by the items of the dimensions "Uncertainty about working conditions" and "Uncertainty about the job" - which

showed a higher coefficient than the Spanish and the Danish questionnaires. The alphas of the other scales are of similar magnitude in all three versions.

It is considered that the results of this study ratify the hypothesis of the theoretical model that bases the COPSOQ, as well as other versions of the questionnaire that have been adapted for other languages and cultures [3, 15, 18–25, 28], and indicate that the final model of this study can be applied in Brazil, considering workers from universities between the ages of 19 and 70 and with full high school education.

The results of using this instrument to assess psychosocial risk in the workplace in common with other countries can allow the standardization of measures; the comparison with reference values in various professional categories; the implementation of appropriate interventions (including preventive interventions); and the organization to monitor multidimensional aspects of work, as it is already happening in Spain [19].

Limitations of this study include: (1) sampling cannot be considered representative of the diversity of the population of Brazilian workers, (2) the online data collection procedure may have presented a response rate below that based on answers written on paper, and (3) some of the responses may have been influenced by Brazil's current economic crisis. The presentation of the limitations of this study may contribute and guide future research on the subject.

The analysis of the model of this study resulted in a medium version of the COPSOQ, consisting of 13 dimensions and 70 items. The results support the final model of this study acceptable levels of reliability and internal validity for the application in Brazil.

It is considered that this medium version may constitute a further step in a broader approach for the final validation of this questionnaire for Brazil, which could be called the COPSOQ II - Brazil medium version.

In future studies, the use of COPSOQ in other contexts of work is suggested, enabling an expansion of the database for analysis, making possible the development of Brazilian standards by category and region of the country, reflecting the continental characteristics of Brazil.

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Conflict of interest

None to report.

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